

- 2 Corals grow in shallow seawater. Corals consist of colonies of small animals called polyps. These polyps have photosynthetic protocists called algae inside their cells, which is advantageous both to the coral polyps and to the algae.

The algae that live within the cells of the polyps can also live independently as free-living algae.

- (a) The rate of photosynthesis of algae that live within the cells of coral polyps is higher than that of free-living algae.

Suggest **and** explain how living inside the cells of coral polyps increases the rate of photosynthesis in these algae compared to free-living algae.

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- (b) The relative abundance of five different chloroplast pigments in the algae of corals was determined. The results are shown in Table 2.1.

**Table 2.1**

| chloroplast pigment | percentage of total |
|---------------------|---------------------|
| chlorophyll a       | 39                  |
| peridinin           | 39                  |
| chlorophyll c2      | 13                  |
| dinoxanthin         | 7                   |
| β-carotene          | 2                   |

Outline the method you would use to separate and identify the pigments present in an extract of these algae.

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- (c) Table 2.2 shows the light wavelengths at which each algal chloroplast pigment shows its two largest peaks of light absorption.

**Table 2.2**

| chloroplast pigment | peak 1 wavelength /nm | peak 2 wavelength /nm |
|---------------------|-----------------------|-----------------------|
| chlorophyll a       | 430                   | 662                   |
| peridinin           | 456                   | 485                   |
| chlorophyll c2      | 450                   | 396                   |
| dinoxanthin         | 442                   | 471                   |
| $\beta$ -carotene   | 454                   | 480                   |

Corals kept in tanks are often illuminated by lamps radiating mostly violet and blue light with wavelengths in the range of 400–490 nm.

With reference to Table 2.1 and Table 2.2, suggest why lamps radiating mostly violet and blue light are expected to increase coral growth.

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